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Wednesday 14 September 2018
1 hour

Additional materials: Multiple Choice Answer Sheet

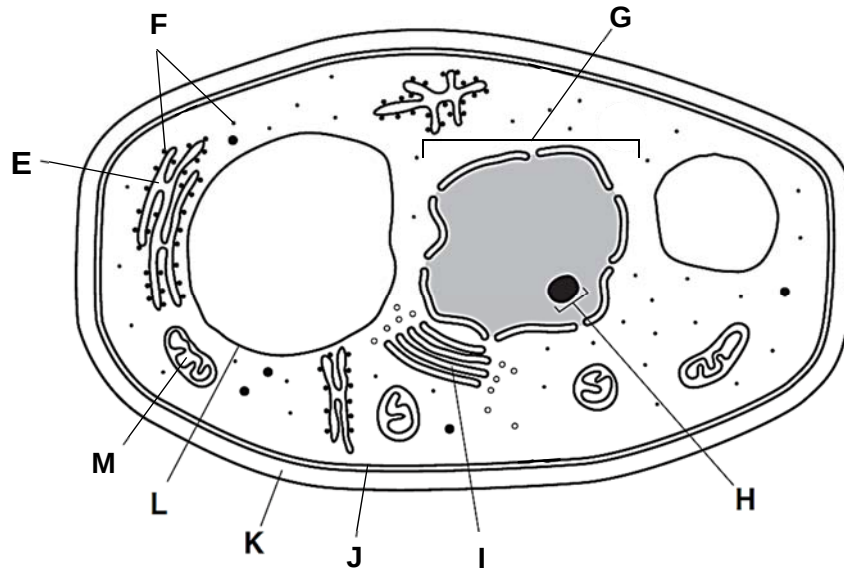
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Tuberculosis and candidiasis are two opportunistic infections that may develop during AIDS. Candidiasis is caused by *Candida albicans*, a yeast-like fungus that lives in human lungs. The figure below shows the structure of *Candida*.

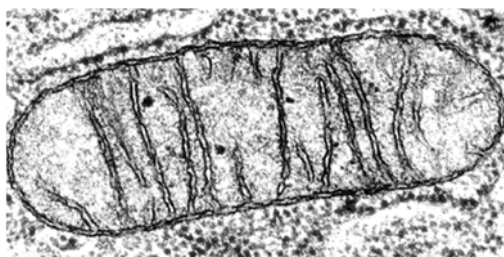


Which of the structure(s) can also be found in the causative agent that causes tuberculosis?

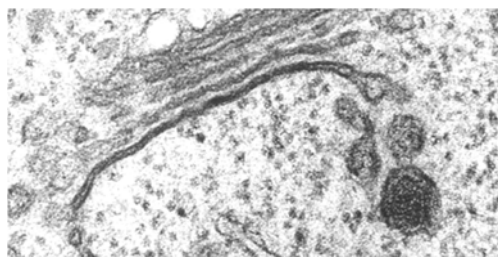
- A None
- B F only
- C F, J, K only**
- D H, J, K only

2 The images below show the electron micrographs of some organelles found in eukaryotic cells.

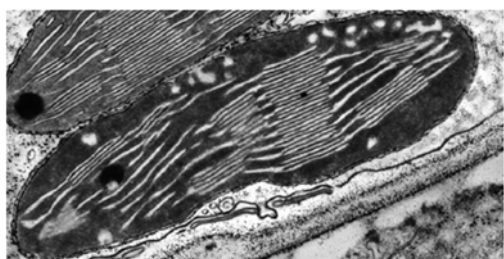
P



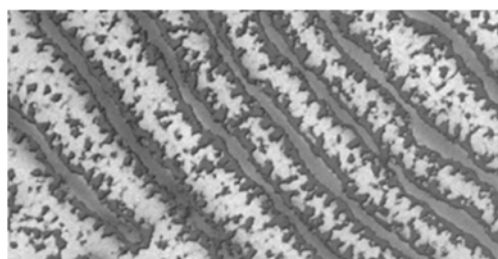
Q



R



S



The following statements are descriptions of membranous cell structures.

- 1 formed by a single membrane and enclosing a large fluid-filled space and regulating the osmotic pressure of the cell
- 2 formed by a single membrane and enclosing inactivated enzymes
- 3 formed by a single membrane that has flattened sacs and tubular structures interconnected throughout the cell, sometimes with a complex of nucleic acid and protein attached
- 4 formed by a single membrane that has tubular structures and containing enzymes to add carbohydrate side chains to proteins
- 5 formed by two membranes and internal membranes that contain pigments
- 6 formed by two membranes whereby the inner membrane is folded extensively
- 7 formed by two membranes, the outer membrane is continuous with another membranous organelle

Which of the following row correctly matches the descriptions of the cell structures?

	P	Q	R	S
A	5	3	6	1
B	5	2	4	7
C	6	4	5	3
D	7	1	2	6

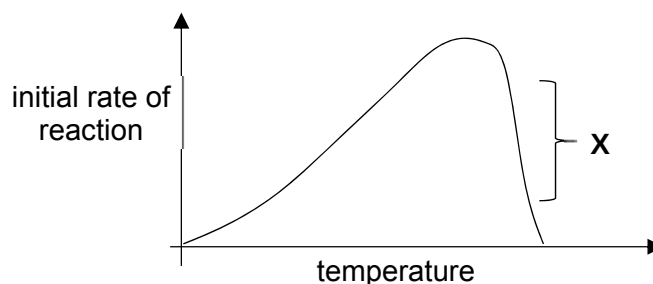
- 3 Particular biological molecules react with chemicals called reagents to give distinct colour changes. The colour depends on the kind of biological molecule and the type of reagent used, as shown in the following table.

chemical reagent	biological molecule	colour change observed
L	protein	violet
M	lipid	red
N	nucleic acid	green

A researcher added different reagents to some isolated ribosomes.

The colour change observed are

- A green only.
 - B red and green.
 - C green and violet.**
 - D violet, red and green.
- 4 The diagram shows the initial rate of reaction using constant amounts of substrate and enzyme at different temperatures.



What is the reason for the decline in the level of activity in region X?

- A breaking of sulphur bridges and ionic bonds in the enzymes**
- B competition between substrate and product for the active site
- C breaking of hydrogen bonds and hydrolysis of peptide bonds in the enzyme
- D insufficient substrates to occupy all the active sites

- 5 Proteins in the cell surface membranes of human cells and mouse cells were labelled with red and green fluorescent dyes respectively.

When a human cell and a mouse cell were fused together, the red and green fluorescent dyes were at first found in different regions of the cell surface membrane of the hybrid cell, but after 40 minutes, they were evenly distributed in the entire cell surface membrane.

What explains this observation?

- A All protein molecules in the cell surface membrane are fixed to structures within the cell, but phospholipid molecules move freely between them.
- B Groups of protein and phospholipid molecules in the cell surface membrane are attached to each other and move together.
- C Only protein molecules in the outer layer of the cell surface membrane can move freely between phospholipid molecules.
- D Protein molecules in the outer layer of the cell surface membrane and those which span the bilayer can move freely between phospholipid molecules.**

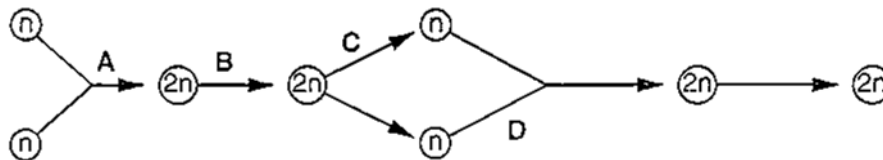
- 6 At prophase of mitosis, a eukaryote chromosome consists of two chromatids.

What is the structure of a single chromatid?

- A one molecule of single-stranded DNA coiled around protein molecules
- B two molecules of single-stranded DNA each coiled around protein molecules
- C one double helix of DNA coiled around protein molecules**
- D two double helices of DNA each coiled around protein molecules

- 7 The diagram represents the life cycle of an animal.

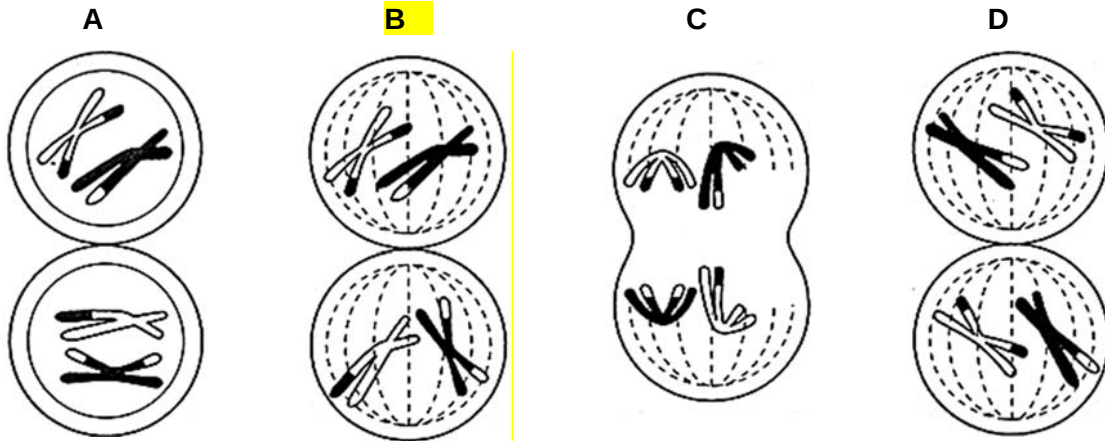
At which stage in the life cycle does mitosis occur?



- 8 The diagram shows anaphase I of meiosis.



Which diagram shows metaphase II as meiosis continues in this cell?



- 9 Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine and the blood.

However, within their own environments, a blood cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a blood cell.

Which statement explains this?

- A Different stem cells have only the genes required for their particular cell line.
- B Genes not required for the differentiation of a particular cell line are methylated.**
- C Binding of repressor molecules prevents the expression of genes not required for a particular cell line.
- D Expression of gene not required for a particular cell line is controlled at translational level.

- 10 The table shows the mode of action of two antibacterial drugs that can affect the synthesis of proteins.

antibacterial drug	rifampicin	streptomycin
mode of action	binds to RNA polymerase	causes errors in translation

If bacteria are treated with both drugs, what will be the immediate effects?

- 1 Transcription will stop, but non-functional proteins may continue to be synthesised.
- 2 If translation has started, proteins may be non-functional.
- 3 Translation will be inhibited.

- A 1, 2 and 3
B 1 and 2 only
 C 1 and 3 only
 D 2 and 3 only

- 11 A peptide consists of ten amino acids of four different kinds.

What is the theoretical minimum number of different kinds of tRNA molecules required to translate the mRNA for this peptide?

- A 4**
 B 10
 C 12
 D 30

- 12 The *trp* operon is an example of which type of transcriptional regulation?

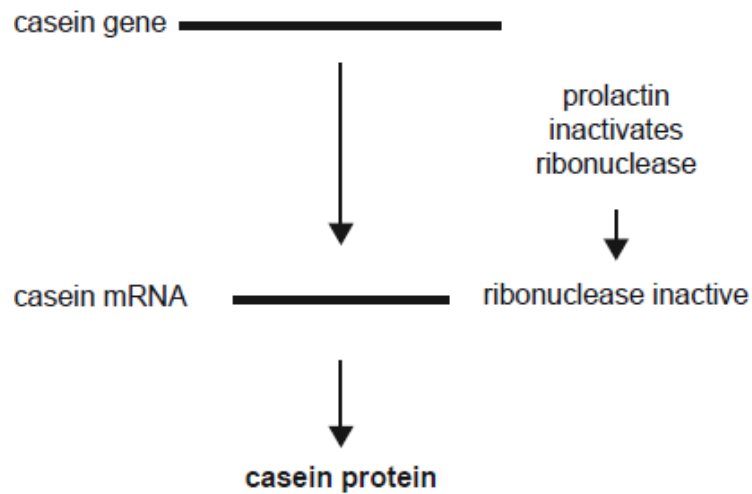
- A A repressor is inactivated by the presence of the amino acid tryptophan
 B An activator is activated by the presence of the amino acid tryptophan
C A repressor is activated by the presence of the amino acid tryptophan
 D An activator is inactivated by the presence of the amino acid tryptophan

- 13 Which of the following statements correctly describes telomerase?

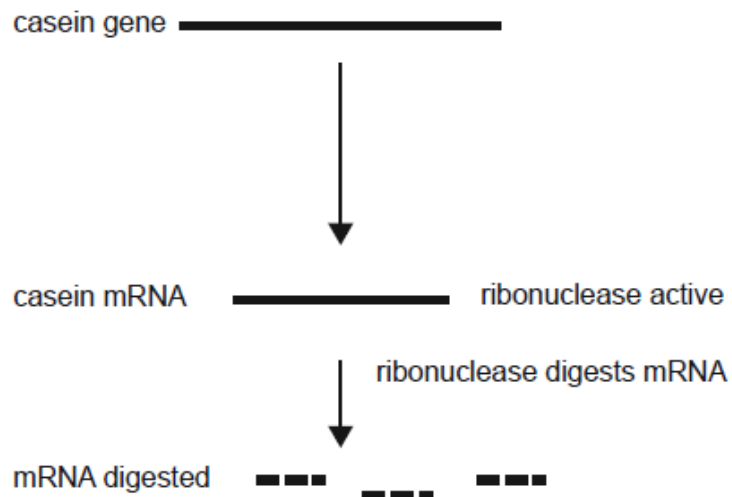
- A Telomerase carries its own DNA template.
 B Inactivation of telomerase contributes to the extended life span of cancer cells.
C Telomerase extends the 3' ends of the parental strand of linear chromosomes.
 D Telomerase extends the 3' ends of the daughter strand of linear chromosomes.

- 14 Casein is a major protein found in mammalian milk.

When the mammals are producing milk, the pathway for the production of casein can be represented as shown in the diagram below.



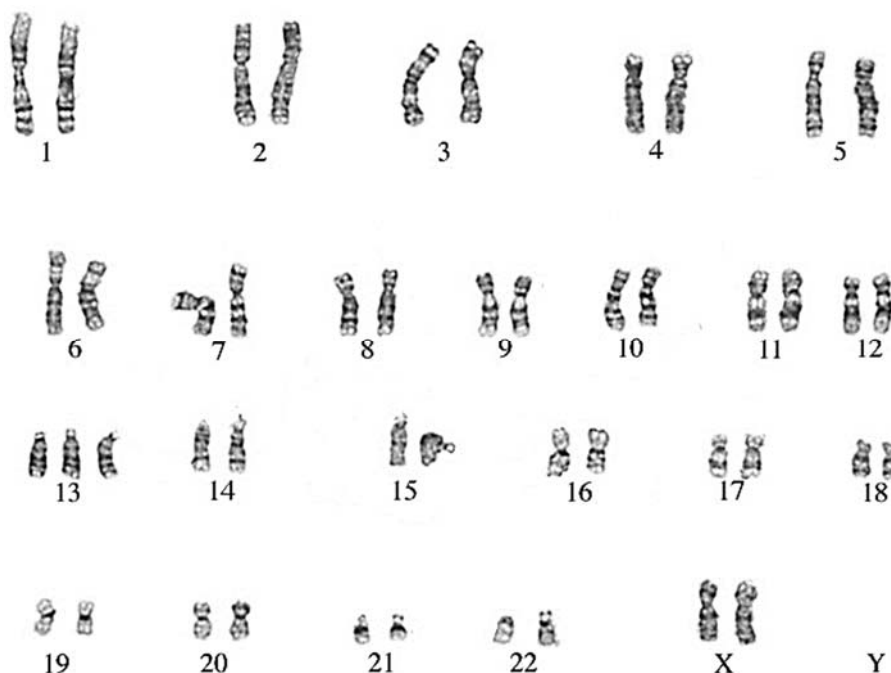
When the mammals are not producing milk, the pathway can be represented as shown in the diagram below.



Which one of the following conclusions can be made from the information above?

- A Ribonuclease has the effect of turning on the casein gene.
- B Casein is a repressor protein for milk production in mammals.
- C The hormone prolactin allows for the expression of the casein gene.**
- D Mammals produce milk only in the absence of the hormone prolactin.

- 15 A newborn baby was diagnosed with Patau syndrome. The diagram below shows her chromosomes.



This is an example of

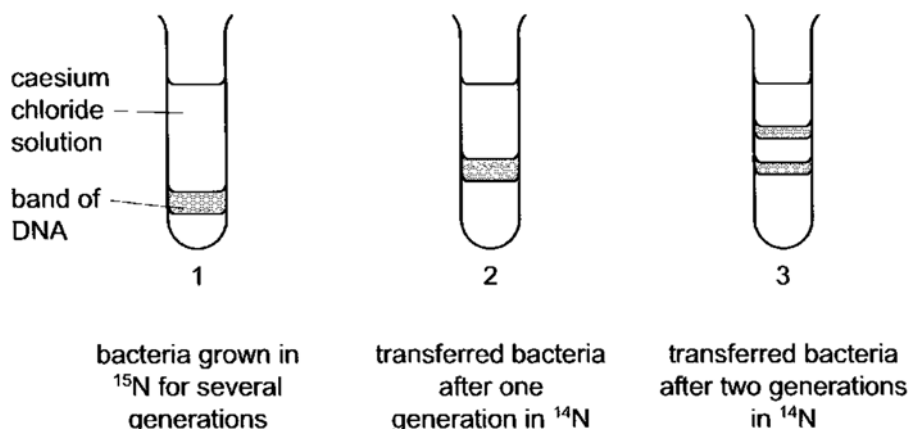
- A frameshift mutation
 B silent mutation
C aneuploidy
 D polyploidy
- 16 Carcinogen **W** can cause changes in tumour suppressor genes, **X**. This can lead to uncontrolled division and the formation of a tumour which may spread to other parts of the body via process **Y**.

Which of the following responses correctly identifies **W**, **X** and **Y**?

	W	X	Y
A	nicotine	<i>ras</i>	mutations
B	asbestos	p53	metastasis
C	tar	<i>ras</i>	metastasis
D	ethanol	<i>p53</i>	mutations

- 17 Bacteria grown in ^{15}N for many generations were transferred to ^{14}N for further replication. DNA from the bacteria was extracted and separated by density gradient centrifugation.

Their results are summarised in the following diagram.



Which of the following process accounts the different positions of the DNA band at different generations?

- A binary fusion
- B binary fission**
- C transformation
- D mitosis
- 18 In rabbits, the color of body fat is controlled by a single gene with two alleles. The outcome of this trait is affected by the diet of the rabbit.

When raised on a standard vegetarian diet, the dominant allele confers white body fat, and the recessive allele confers yellow body fat.

However, when raised on a xanthophyll-free diet, the homozygous recessive animal has white body fat.

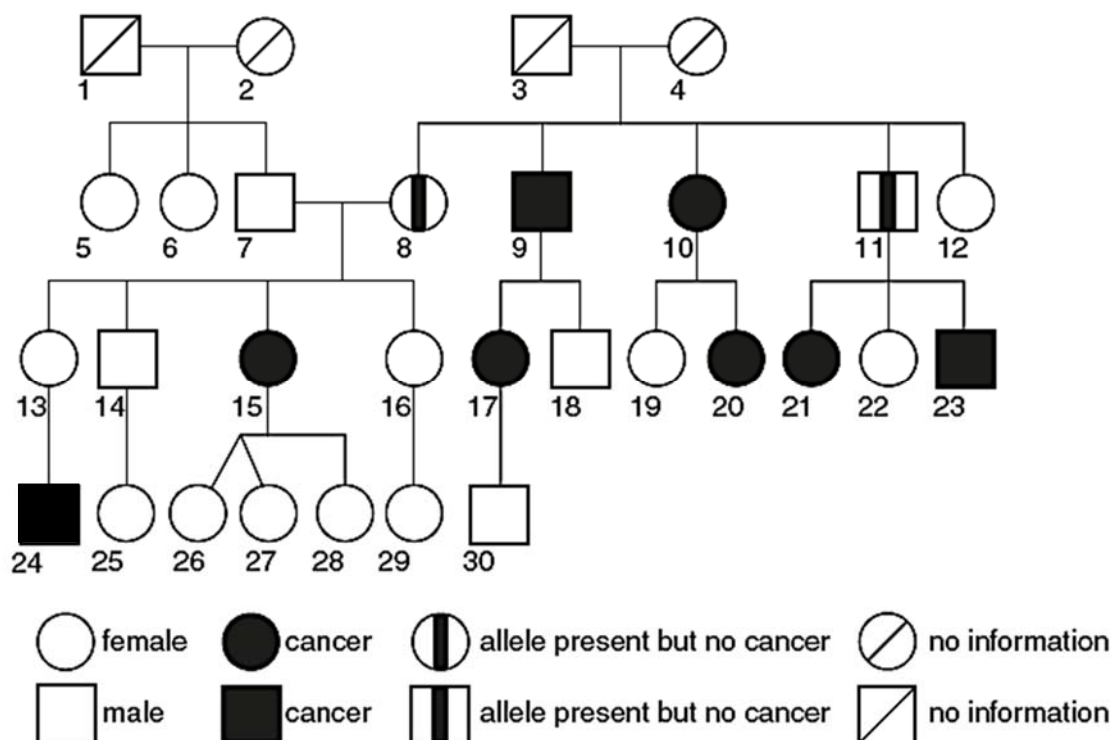
If a heterozygous animal is crossed to a rabbit with yellow body fat, what are the proportions of offspring with white and yellow body fat?

	raised on standard vegetarian diet	raised on xanthophyll-free vegetarian diet
A	1 white body fat : 1 yellow body fat	All white body fat
B	1 white body fat : 1 yellow body fat	1 white body fat : 1 yellow body fat
C	3 white body fat : 1 yellow body fat	1 white body fat : 1 yellow body fat
D	3 white body fat : 1 yellow body fat	3 white body fat : 1 yellow body fat

- 19 The BRCA2 protein is involved in suppressing the development of tumours. The gene that codes for this protein is on chromosome 13.

Several different dominant alleles of this gene, *BRCA2*, code for faulty versions of the protein. The presence of any one of these faulty alleles leads to an increased chance of developing several types of cancer, including breast cancer. Not everyone with one of these alleles develops cancer.

The pedigree (family tree) below shows the occurrence of cancers in four generations of a family. The presence of a faulty *BRCA2* allele was confirmed in person 15. The other individuals with cancer were not tested for the presence of the allele. For individuals 17 to 30, only one of their parents is shown in the pedigree. Individuals 24–30 are all under twelve years old.



Which one of the following statement is **not** correct?

- A Individuals 8 and 11 have *BRCA2* allele and may develop cancer later in life.
- B Individuals 8 to 11 may have inherited *BRCA2* allele from either of their parents.
- C Individual 15 may have inherited one copy of *BRCA2* allele from her mother.
- D Individual 24 may have inherited the *BRCA2* allele only from his mother and not his father.**

20 The table below shows the results of the stomatal count investigation.

	number of stomata visible at $\times 100$ magnification										mean number of stomata
sun leaves	21	24	36	24	15	18	27	33	18	24	24 ± 6
shade leaves	36	39	35	42	28	36	34	40	48	32	37 ± 6

The table below shows the critical values for a two-tailed t -test, where probability < 0.05 .

degrees of freedom	8	9	10	18	19	20
probability 0.05	2.306	2.262	2.228	2.101	2.093	2.086

What conclusion can be made between the mean number of stomata of sun leaves and shade leaves?

- A The calculated t -test value of 4.743 is greater than the critical t -test value of 2.306, hence the mean number of stomata of sun leaves is different from shade leaves.
- B The calculated t -test value of 4.743 is greater than the critical t -test value of 2.262, hence the mean number of stomata of sun leaves is less than shade leaves.
- C The calculated t -test value of 4.743 is greater than the critical t -test value of 2.101, hence the mean number of stomata of sun leaves is different from shade leaves.
- D The calculated t -test value of 4.743 is greater than the critical t -test value of 2.093, hence the mean number of stomata of sun leaves is less than shade leaves.

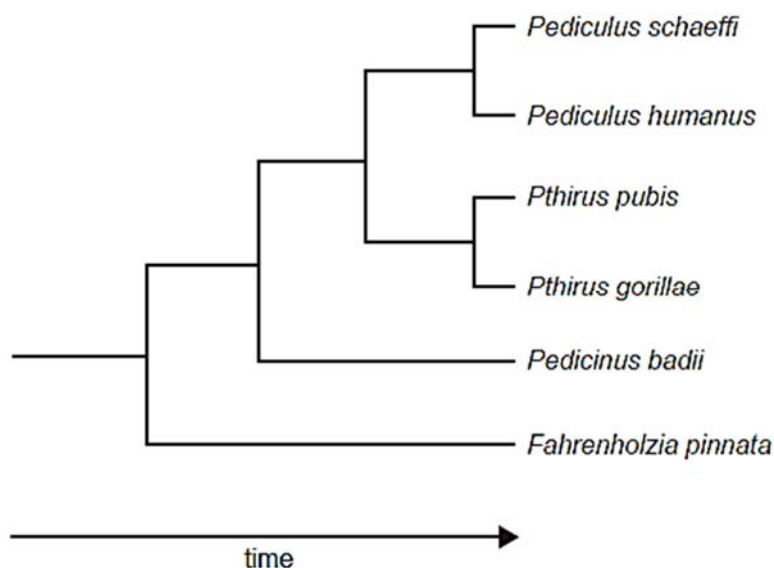
21 Northern elephant seals, *Mirounga angustirostris*, were nearly hunted to extinction in the 1890s, with only about 20 individuals left at the end of the century. The population has now grown to more than 120 000.

In the 1890s, southern elephant seals, *Mirounga leonina*, were not as severely hunted and currently there are estimated to be 600 000 southern elephant seals.

Based on this information, it is true to say that

- A northern elephant seals have evolved as a result of the 'founder effect'.
- B northern elephant seals would show less genetic variation than southern elephant seals.
- C southern elephant seals would have experienced greater genetic drift than northern elephant seals.
- D the mutation rate in northern elephant seals would have been greater than in southern elephant seals.

- 22 The phylogenetic tree for different species of lice is shown below. The tree has been constructed based on molecular and morphological data.



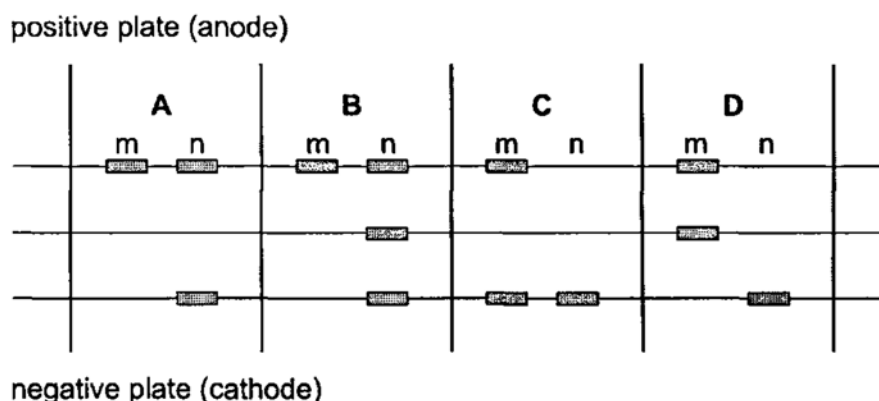
This information suggests that

- A** *Pedicinus badii* shares a more recent common ancestor with *Pthirus gorillae* than with *Fahrenholzia pinnata*.
- B** *Pediculus humanus* is more closely related to *Pedicinus badii* than it is to *Pthirus pubis*.
- C** six species of lice have evolved by convergent evolution.
- D** *Pediculus schaeffi* is the ancestor of *Pediculus humanus*.
- 23 There are two forms of the seeds of the garden pea, *Pisum sativum*, one with a wrinkled skin and the other with a smooth skin.

Plants with wrinkled seeds have a recessive mutation caused by the insertion of an extra nucleotide sequence into a gene.

Two heterozygous pea plants were crossed. Two of the offspring (m and n) had the DNA for this gene removed and separated by electrophoresis.

Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?

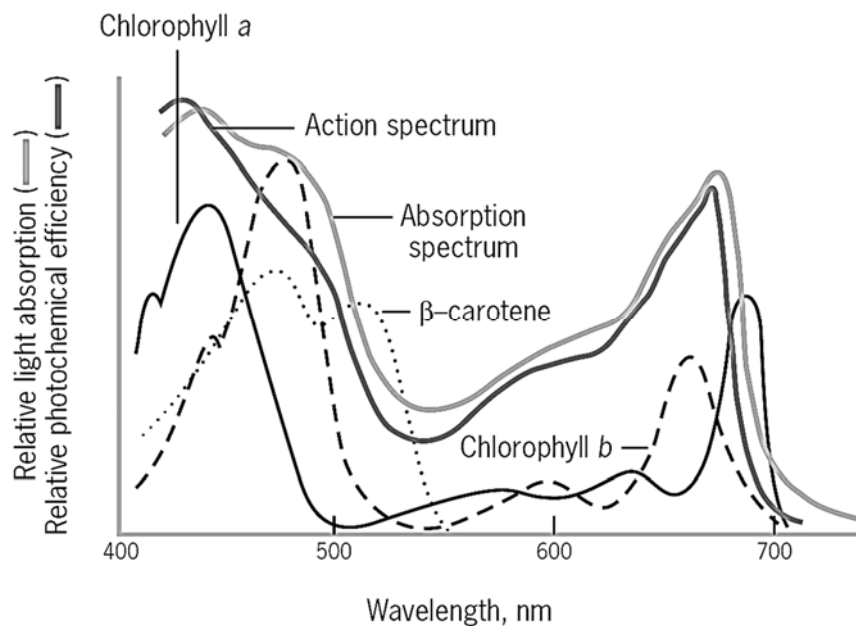


- 24 During aerobic respiration ATP can be formed by glycolysis and oxidative phosphorylation in the electron transport system.

In the complete oxidation of one molecule of glucose, approximately what percentage of ATP is formed by oxidative phosphorylation?

- A 10%
- B 25%
- C 75%
- D 90%**

- 25 The figure below shows the absorption spectrum of the photosynthetic pigments of a flowering plant and its action spectrum.



What can be concluded from the graph above?

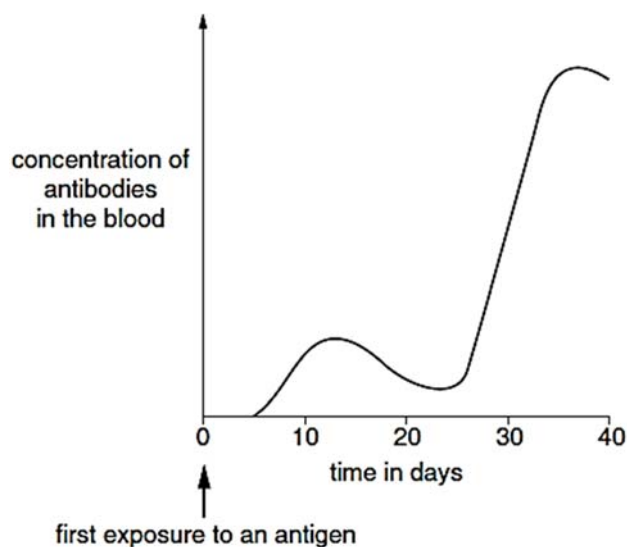
- 1 The relative light absorption will be higher at higher temperatures, as temperature is a limiting factor.
- 2 The green leaves reflect light of wavelength 550 nm, hence the photochemical efficiency is low.
- 3 The compensation point of β -carotene, whereby the rate of photosynthesis equals the rate of respiration, occurs at 550nm.
- 4 The accessory pigments chlorophyll b and β -carotene absorb light energy mostly at 480nm.

- A 2 and 4**
- B 1, 2 and 3
- C 1, 3 and 4
- D All of the above

26 Which of the following statement is true?

	blood glucose concentration	hormone secreted	receptor involved at target cell	cellular effect
A	increase above normal level	insulin	receptor tyrosine kinase	decrease in blood glucose concentration
B	decrease below normal level	glucagon	G-protein coupled receptor	decrease in blood glucose concentration
C	decrease below normal level	insulin	G-protein coupled receptor	increase in blood glucose concentration
D	increase above normal level	glucagon	receptor tyrosine kinase	increase in blood glucose concentration

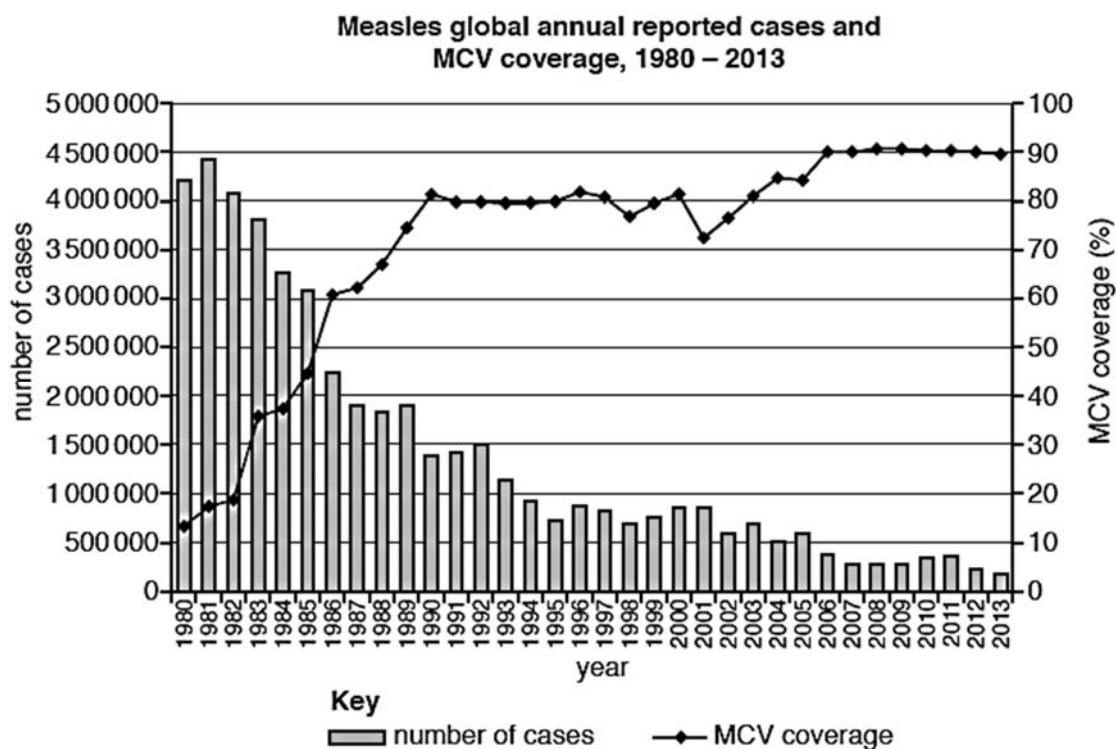
27 The graph shows the amount of antibody produced in response to an antigen.



From the graph, which statement is correct?

- A It takes 35 days to achieve active immunity.
- B Memory cells for this antigen are present in the body within 20 days.**
- C A second exposure to the antigen occurred on day 20.
- D T helper cells are activated on day 12.

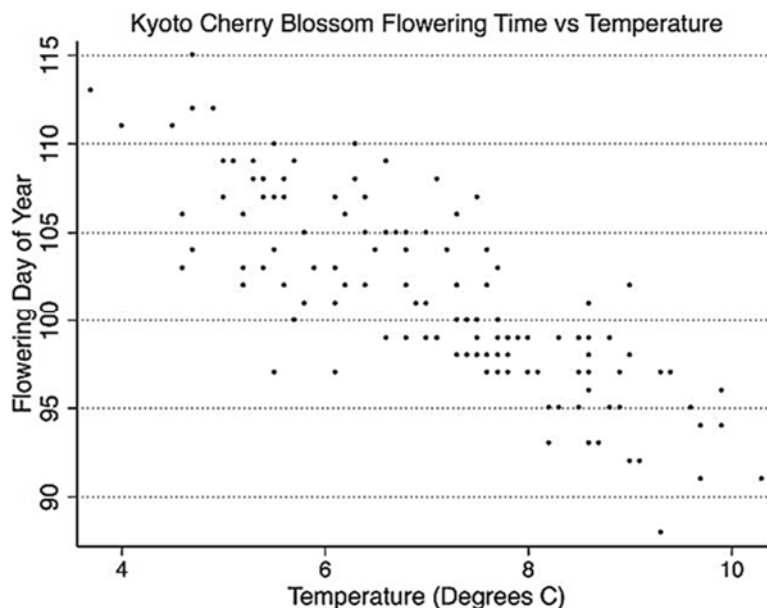
- 28 Measles is a serious disease that can be prevented by vaccination. The chart below shows the Measles-containing Vaccine (MCV) coverage and annual reported cases of measles between 1980 and 2013.



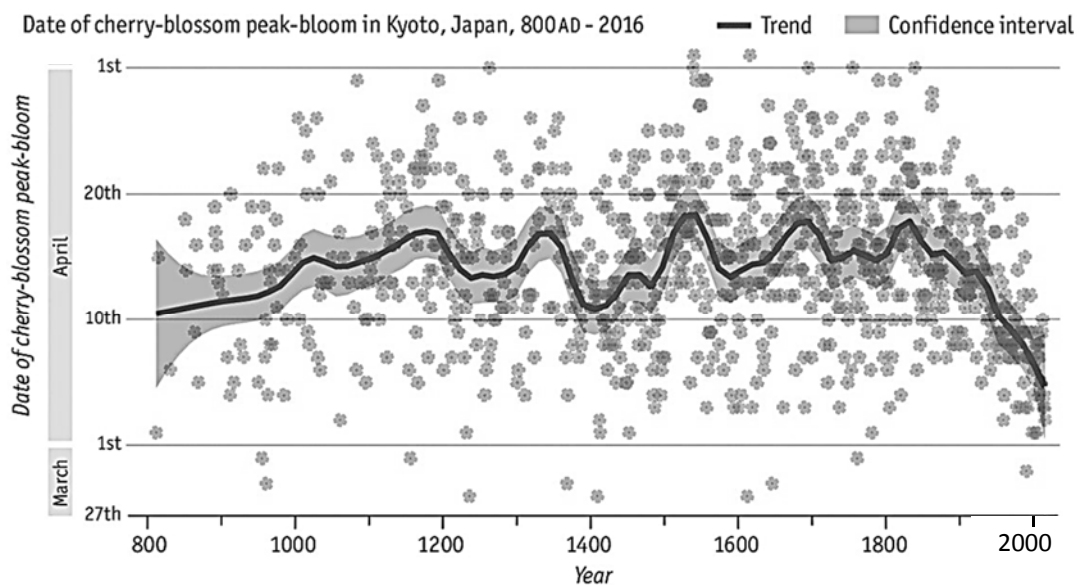
Which of the following statements is a correct interpretation of the chart?

- A The highest number of measles cases occurred when MCV coverage was at its lowest.
- B An increase in herd immunity resulted in fewer deaths from measles.**
- C A 80% MCV coverage resulted in fewer than half a million cases of measles each year.
- D There is a positive correlation between the number of measles cases and the MCV coverage.

29 The effect of temperature on cherry blossom flowering time (day of the year) is shown below.



The records of timing of cherry blossoms in Japan from 800 A.D is shown below.



Which conclusions can be made from both graphs?

- 1 The peak of cherry blossoms has consistently been earlier since 1850. Cherry blossoms begin earlier as temperature increases from 4 to 10°C.
- 2 The temperature in Japan has been increasing since 800 A.D., resulting in later blooming and pollinators are unable to pollinate the cherry trees.
- 3 No conclusion can be made as the data points are scattered and lack clear trend.

A 1 only

B 2 only

C 3 only

D 1 and 2

- 30 An investigation was carried out to assess the effect of diet on the milk yield and methane production of cows. The cows in group **A** were fed a traditional diet and those in group **B** were fed the same diet with a mixture of chopped hay and straw added.

The table below shows the results of this investigation.

Group	Mean milk yield per cow/ $\text{dm}^3 \text{ day}^{-1}$	Methane emission for each dm^3 milk produced / dm^3
A	24.0	30.0
B	27.6	24.0

Which of the following actions will help reduce the impact of global warming?

- 1 Decreasing consumption of beef and milk
- 2 Creating more foraging grounds to feed the cows
- 3 Adding chopped hay and straw to the cows' diet.

- A** 1 only
B 1 and 3 only
C 2 and 3 only
D All of the above